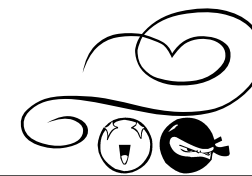


combinatorics

Part A: Permutations



A1	BAULKHAM HILLS	☒	☐
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Find the number of ways all the letters of the word EPSILON can be arranged such that the three vowels are all next to each other. 1

- (A) 7! (B) 5! (C) 4!5! (D) 3!5!

three vowels grouped together = 3!
Group of vowels plus four consonants = 5! $\therefore$ total = 3!5!

A2	BAULKHAM HILLS	☒	☐
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Five digit numbers are to be formed using the digits 5, 6, 7, 8 and 9. 3
How many odd numbers greater than 70000 can be formed if no digit is repeated?

restrictions * first digit must be 7, 8 or 9
* last number must be 5, 7 or 9
Case 1: first digit 7 or 9 Ways = $2 \times 2 \times 3! = 24$
Case 2: first digit is 8 Ways = $1 \times 3 \times 3! = 18$
2 ways of first number 1 way of first number
Leaving 2 numbers for last spot Leaving 3 numbers for last spot
Arrange the 3 numbers in the middle Arrange the 3 numbers in the middle
Total odd numbers > 70000 = $24 + 18 = 42$

A3	BAULKHAM HILLS	☒	☐
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At a family gathering, 6 adults and 3 children are randomly seated around a circular table. 2

What is the number of possible seating arrangements if none of the children are to sit together?

- (A) 720 (B) 2400 (C) 14400 (D) 86400

Arrange adults in circle $5!$
Place 3 children into the 6 spaces between adults $6P_3$ or $6 \times 5 \times 4$
 $\therefore$ No. of ways = $5! \times 6P_3 = 14400$

A4	BAULKHAM HILLS	☒	☐
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All the letters of the word DEFINITION are arranged in a row.

- (i) How many distinct arrangements are possible? 1
(ii) In how many of these arrangements are all the vowels together? 2
(iii) What is the probability that the letter D appears somewhere to the left of the letter T? 1

d) DEFINITION
10 letters
5 vowels
5 consonants
(i) $\frac{10!}{2!2!} = 302400$ answer
(ii) Arrange consonants $\frac{5!}{2!} = 60$
(D, F, N, T, M)
Place the vowel group in one of the 6 places indicated
Arrange the vowels within their group $\frac{5!}{2!}$
 $\therefore 60 \times 6 \times \frac{5!}{2!} = 60 \times 6 \times 20 = 7200$

(iii) The letter D can either appear to the left or to the right of the letter N, with equal probabilities.
 $\therefore$ Probability = $\frac{1}{2}$

A5	CARINGBAH	☒	☐
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Find the number of ways in which 2 boys and 5 girls can be arranged in a circle if:

- (i) there is no restriction. 1
(ii) the boys and girls are in separate groups. 1
(iii) the boys are separated. 2

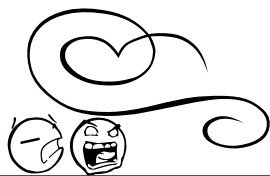
(i) Fix one person, rearrange 6 people So $6! = 720$.
(ii) Fix one girl, 2! ways to rearrange the boys, 5! ways to rearrange the remaining 4 girls & 1 group of 2 boys, Total ways = $5! \times 2! = 240$
(iii) 2 boys separate = total - 2 boys separate = $6! - 5! \times 2! = 480$

A6	INDEPENDENT	☒	☐
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Find the number of ways the letters of the word TRAPEZIUM can be arranged in a line

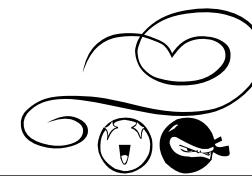
- (i) without restriction. 1
(ii) so that no two consonants are next to each other. 2

a.i. Outcomes assessed: ME11-5
Answer $9! = 362880$ arrangements
a.ii. Outcomes assessed: ME11-5
Answer
Pattern must be CVCVCVCVCVC Number of arrangements is $5! \times 4! = 2880$



combinatorics

Part A: Permutations



Unit

NAME: _____

A7	CARINGBAH	☒	☐
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Three girls and three boys sit around a round table.

What is the probability that they sit boy-girl-boy-girl-boy-girl?

$$P(\text{Boy - girl - boy - girl}) = \frac{3!2!}{5!} = \frac{1}{10}$$

- (A) $\frac{1}{10}$ (B) $\frac{3}{20}$ (C) $\frac{1}{120}$ (D) $\frac{1}{20}$

A8	CARINGBAH	☒	☐
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Find the number of ways in which the letters of the word TRIANGLE can be arranged in a line:

- i) so that all the vowels are next to each other?
ii) so that consonants are in the two end positions?

- i) all the vowels next to each other:
 $2! \times 3! \times 5! = 1440$
ii) consonants are in the two end positions:
Choose the first and last letters from the 5 consonants, then arrange the remaining 6 letters between them in $(5 \times 4) \times 6! = 14400$

A10	INDEPENDENT	☒	☐
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Find the number of ways in which the letters of the word HEXAGON can be arranged in a straight line

- (a) so that the vowels are all next to each other.
(b) so that no two consonants are next to each other

a. Outcomes assessed: ME11-5, -7

Answer

Arrange (E, A, O), H, X, G, N in 5! ways, then arrange the vowels in 3! ways. $\therefore 5! \times 3! = 720$ ways

b. Outcomes assessed: ME11-5, -7

Answer

Only possible pattern is C V C V C V C where C is a consonant and V is a vowel.
Arrange the consonants in 4! ways and the vowels in 3! ways $\therefore 4! \times 3! = 144$ ways

A11	INDEPENDENT	☒	☐
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In how many ways can a group of 6 people line up in a queue so that the oldest person is in front of the youngest person?

- (A) 180 (B) 360 (C) 540 (D) 720
 $\frac{1}{2} \times 6! = 360$

A12	HURLSTONE	☒	☐
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How many unique arrangements of the letters in the word MATHEMATICS are possible?

- A. 11! B. $\frac{11!}{6}$ C. $\frac{11!}{8}$ D. 8!

A13	HURLSTONE	☒	☐
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How many unique arrangements of the letters in the word SWANS are possible, if the two S's are not next to each other?

There are 6 cases where the two S's are not alongside each other. Each case can have the other letters arranged in 3! ways.

$$\text{Solution} = 6 \times 3! = 36$$

Alternatively: (Total possible arrangements of 5 objects minus those with two objects adjacent) divided by 2 gives:

$$(5! - 4! \times 2!) \div 2 = 36$$

A15	JAMES RUSE	☒	☐
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At a particular dinner, each rectangular table has nine seats, five facing the stage and four with their backs to the stage.

- (i) In how many ways can 9 people be seated at the table if John and Mary sit on the same side?
(ii) What is the probability of John and Mary sitting on opposite sides of the table?

d) facing stage:
J has 5 to choose from
M has 4 to choose from
backs to stage:
J has 4 to choose from
M has 3 to choose from
Other 7 arranged in 7! ways
 $\therefore$ J & M can be seated in $4 \times 3 \times 7! + 5 \times 4 \times 7!$
 $= 7! (12 + 20)$
 $= 16! 280$ ways

No. of ways J & M can be seated on opposite sides:
or
No. of ways $= 4 \times 5 \times 7! + 4 \times 5 \times 7!$
 $= 40 \times 7! = 201600$
P(J & M on opposite sides)
 $= \frac{201600}{9!} = \frac{5}{9}$

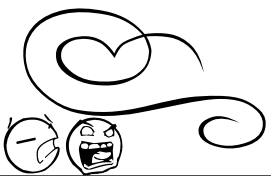
A16	KNOX	☒	☐
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The letters of STREET are arranged randomly in a row. Find the probability that:

- (i) the two letters E are together.
(ii) the Es and Ts are together in one group.

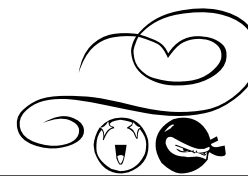
(i) Total arrangements $= \frac{6!}{2!2!} = 180$
(1) 2 Es together:
Consider them as 1 group & from the rest they can be arranged in $\frac{5!}{2!} = 60$
 $\therefore$ Probability $= \frac{60}{180} = \frac{1}{3}$

(ii) Es & Ts are together in one group:
4 letters or one group can be arranged in $\frac{4!}{2!} = 6$ ways
Together with other 2 letters it can be arranged in $3! \times 2! = 12$ ways
 $\therefore$ P = $\frac{12}{180} = \frac{1}{15}$



combinatorics

Part A: Permutations



A25	TOTAL EDUCATION	☒	☐
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Three-digit numbers are formed from the digits 1, 2, 3, 4 and 5 where each digit may only be used once. What is the probability that a particular number formed is divisible by three? **1**

- A. $\frac{1}{5}$ B. $\frac{2}{5}$ C. $\frac{1}{2}$

way with no restriction = $5C_3 = 10$
 # of sums which are multiples of 3 = 4
 from 10 above
 Probability = $\frac{4}{10}$ or $\frac{2}{5}$

A26	TOTAL EDUCATION	☒	☐
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The letters of the word TACTICS are arranged randomly in a row. What is the probability that there are 3 letters between the two T's. **2**

T A C T I C S T S C I T _ _

3!

Remember $2 \times C = 2!$ so $5 \times 4 \times 3 \times 2! = 180$ way

No restriction = $7!$

$\Rightarrow$ Probability = $\frac{180}{7!} = \frac{1}{28}$

A27	TOTAL EDUCATION	☒	☐
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Four girls and three boys are arranged in a circle. How many ways can this occur, if:

- (i) there are no restrictions? **1**
 (ii) a particular boy wishes to sit beside a particular girl? **2**

(i) $\frac{7!}{7} = 6! = 720$ (ii) treat boy and girl as one group
 $\therefore \frac{6! \cdot 2!}{6} = 5! \cdot 2! = 240$

A28	TOTAL EDUCATION	☒	☐
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A group of four engineers and four lawyers is to be selected from six engineers and six lawyers and arranged around a circular table for a meeting. **3**

In how many ways can this be done if Eddie the Engineer is on the table?

With Eddie the Engineer in the group, 4 engineers and 4 lawyers can be selected in $1 \times {}^5C_3 {}^6C_4 = 150$ ways. These 8 people can be seated around a circular table in $7!$ ways.
 Hence, total arrangements = $150 \times 7! = 756000$

A29	TOTAL EDUCATION	☒	☐
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The ten letters of the word FACILITATE are written on separate cards and arranged in a row. **3**

How many different arrangements are possible if all of the vowels are together and do not appear in alphabetical order when viewed in either direction?

F:1 A:2 C:1 I:2 L:1 T:2 E:1

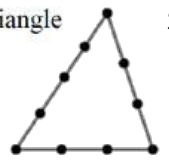
Group vowels together $\rightarrow$ total of 6 objects to arrange in $\frac{6!}{2!}$ ways (taking the duplicate Ts into account)

Group of 5 vowels can be arranged in $\frac{5!}{2!2!}$ ways (taking the duplicate As and Is into account). Out of these arrangements, only two of them will appear in alphabetical order in either direction. Hence, the vowels can be arranged amongst each other (and not in alphabetical order) in $(\frac{5!}{2!2!} - 2)$ ways.

Hence, total arrangement is $\frac{6!}{2!} \times (\frac{5!}{2!2!} - 2) = 10080$

A30	SYDNEY GRAMMAR	☒	☐
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A straight line is to be drawn through two of the twelve points on the triangle shown. How many different straight lines can be drawn? **2**



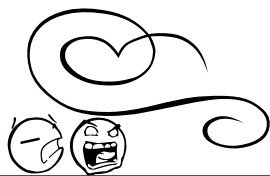
Since there are 12 points on the triangle, there are ${}^{12}C_2$ ways to select 2 points. We remove lines connecting two points on the same side of the triangle, i.e. we subtract 4C_2 , 4C_2 and 4C_2 . We also need to add 3 lines corresponding to each side of the triangle.

$\therefore$ total different lines = ${}^{12}C_2 - {}^4C_2 - {}^4C_2 - {}^4C_2 + 3 = 47$

A31	TRIAL ENTERPRISES	☒	☐
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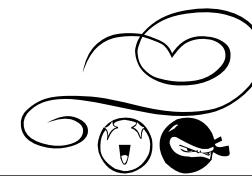
The number of ways the letters from the word 'noon' can be arranged is **1**

- (A) 2 (B) 4 (C) 6 (D) 24



combinatorics

Part A: Permutations



A32	TRIAL ENTERPRISES	☒	☐
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A 10-person committee is to be formed from a group of 12 women and 16 men. In how many ways can the committee be formed if:

- (a) The committee must contain 5 men and 5 women.
(b) There must be at least 5 more women than men.

$${}^{12}C_3 \times {}^{16}C_5 = 792 \times 4368 = 3459456$$

Question 15 (b)

there must be 8, 9 or 10 women

$${}^{12}C_8 \times {}^{16}C_2 + {}^{12}C_9 \times {}^{16}C_1 + {}^{12}C_{10} \\ = 495 \times 120 + 220 \times 16 + 66 = 62986$$

2

2

A33	TRIAL ENTERPRISES	☒	☐
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A bag contains 5 marbles of different colours – green, blue, yellow, orange and black. In how many ways can these five marbles be arranged in a line if the green ball must be first?

1

$$4! = 24$$

A34	TRIAL ENTERPRISES	☒	☐
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In how many ways can five boys and two girls be arranged in a row if the two girls are together?

- (A) 300 (B) 1440 (C) 10 (D) 10080

1

A35	TRIAL ENTERPRISES	☒	☐
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There are two identical black marbles and four identical white marbles arranged in a row.

- (a) How many different arrangements are possible?
(b) How many different arrangements are possible in a row, when using only five marbles.

1

2

Question 9 (b)

$$\frac{6!}{2!4!} = 15 \quad 4W \ 1B = \frac{5!}{4!} = 5 \\ 3W \ 2B = \frac{5!}{3!2!} = 10 \\ \text{Total number} = 15$$

A36	MISC.	☒	☐
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3 boys and 3 girls are to be arranged around a circular table so the boys and girls alternate. How many unique arrangements are possible?

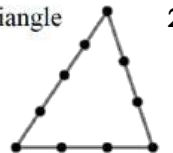
1

- A. 6! B. $3! \times 2!$ C. $3! \times 3!$ D. 5!

A30	SYDNEY GRAMMAR	☒	☐
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A straight line is to be drawn through two of the twelve points on the triangle shown. How many different straight lines can be drawn?

2



Since there are 12 points on the triangle, there are ${}^{12}C_2$ ways to select 2 points. We remove lines connecting two points on the same side of the triangle, i.e. we subtract 4C_2 , 4C_2 and 4C_2 . We also need to add 3 lines corresponding to each side of the triangle.

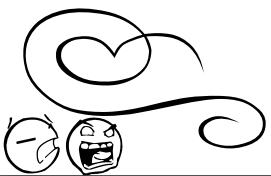
$$\therefore \text{total different lines} = {}^{12}C_2 - {}^4C_2 - {}^4C_2 - {}^4C_2 + 3 = 47$$

A31	TRIAL ENTERPRISES	☒	☐
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The number of ways the letters from the word 'noon' can be arranged is

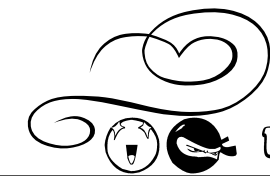
1

- (A) 2 (B) 4 (C) 6 (D) 24



combinatorics

Part B : combinations



B1	ASCHAM	☒	☐
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A group of 12 students is to be divided into discussion groups.

(i) In how many ways can the discussion groups be formed if there are 7 people in one group and 5 people in the other? 1

(ii) In how many ways can the discussion groups be formed if there are 3 groups containing 4 people each? 2

(iii) In how many ways can a president and vice president be chosen for the group of 12? 1

(i) ${}^{12}C_7 \times {}^5C_5$ (or just ${}^{12}C_7$) = 792

(ii) $\frac{{}^{12}C_4 \times {}^8C_4 \times {}^4C_4}{3!} = \frac{34650}{6} = 5775$

(iii) $12 \times 11 = 132$

B2	BONNYRIGG	☒	☐
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A group consists of 5 brother and sister pairs from 5 different families. In how many ways can 2 boys and 2 girls be chosen from the group so as not to include members of the same family? 1

(A) 10 (B) 20 (C) 30 (D) 120

Please check $5C3 \times 3C2 = 10 \times 3$

B3	CANLEY VALE	☒	☐
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In a class there are 25 students, 10 of them are boys and the rest are girls. In how many ways can a committee of 4 students from the class be formed if

i) there is no restriction? 1

ii) at least one boy and one girl are to be included in any committee? 2

i). ${}^{25}C_4 = 12650$

ii). ${}^{10}C_1 \times {}^{15}C_3 + {}^{10}C_2 \times {}^{15}C_2 + {}^{10}C_3 \times {}^{15}C_1 = 11075$

B4	INDEPENDENT	☒	☐
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Find the number of ways that a team of 4 can be chosen from 5 boys and 6 girls

(a) with equal numbers of boys and girls. 1

(b) with a majority of girls. 1

a. Outcomes assessed: ME11-5
Answer ${}^5C_2 \times {}^6C_2 = 150$

b. Outcomes assessed: ME11-5
Answer ${}^6C_4 \times {}^5C_0 + {}^6C_3 \times {}^5C_1 = 115$

B5	HURLSTONE	☒	☐
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Twenty students, ten male and ten female, are travelling from school to the pool for the annual swimming carnival. Twelve of them go in a minibus, six of them on bikes and two of them walk.

(i) Write a mathematical expression that will give the number of ways the students can be divided for the trip. 1

(ii) Show all working to give the numerical answer for the number of ways the students can be distributed, if none of the boys walk. 2

(i) ${}^{20}C_{12} \times {}^8C_6 \times {}^2C_2$ filling minibus, then bikes, then walking.
Other rearrangements are possible.

(ii) Girls walking in ${}^{10}C_2$ ways.
Solution = ${}^{10}C_2 \times {}^{18}C_{12} \times {}^6C_6 = 45 \times 18564 \times 1 = 835380$

B6	HURLSTONE	☒	☐
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A committee of 5 members is to be selected at random from 5 Australians and 3 New Zealanders. Determine the probability that there will be a majority of Australians on the committee. Show all working out. 3

$P(\text{Majority}) = P(3 \text{ or } 4 \text{ or } 5 \text{ Australians})$

$= \frac{{}^5C_3 \times {}^3C_2 + {}^5C_4 \times {}^3C_1 + {}^5C_5}{{}^8C_5} = \frac{30+15+1}{56} = \frac{46}{56}$

B7	CHEL TENHAM GIRLS	☒	☐
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There are 8 girls and 7 boys in a class.

(a) The class elects a captain and vice-captain. In how many ways is this possible? 1

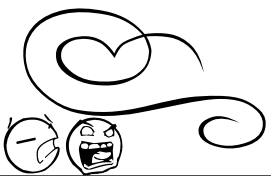
(b) The class elects four representatives for the student council. How many different groups of representatives are possible? 1

(c) How many groups of representatives for the student council are possible, if the class decide that they want two girls and two boys in the group? 1

A) $15 \times 14 = 210$

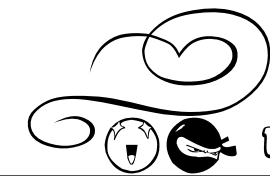
B) ${}^{15}C_4 = 1365$

C) ${}^8C_2 \times {}^7C_2 = 28 \times 21 = 588$



combinatorics

Part B : combinations



Unit

NAMES:

B8	INDEPENDENT	☒	☐
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Find the number of ways that 4 students can be chosen from 6 juniors and 3 seniors

- (i) without restriction. **1**
 (ii) so that there are unequal numbers of juniors and seniors. **2**

a.i. Outcomes assessed: ME11-5
 Answer ${}^9C_4 = 126$
 a.ii. Outcomes assessed: ME11-5
 Answer ${}^9C_4 - {}^6C_2 \times {}^3C_2 = 126 - 45 = 81$

B9	INDEPENDENT	☒	☐
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In how many ways can a group of 3 boys and 3 girls line up in a queue so that the 3 boys are all next to each other and the 3 girls are in alphabetical order? **1**

- (A) 6 (B) 24 (C) 36 (D) 144

Order the girls one way alphabetically. Then 4 positions for group (BBB), with the boys arranged in 3! ways. $\therefore 4 \times 3! = 24$ arrangements.

B10	INDEPENDENT	☒	☐
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Find the number of ways that a team of 7 players can be chosen from 4 teachers and 6 students so that the team contains at least 2 teachers and more students than teachers. **2**

T	S	Ways
2	5	${}^4C_2 \times {}^6C_5 = 36$
3	4	${}^4C_3 \times {}^6C_4 = 60$

$\therefore 96$ ways

B11	CARINGBAH	☒	☐
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Find the number of ways in which four numerals can be chosen from the numerals 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition so that

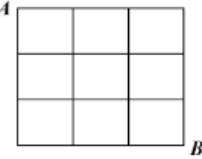
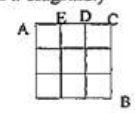
- (i) no odd numerals are chosen. **1**
 (ii) at least one odd numeral is chosen. **1**
 (iii) a majority of odd numerals is chosen. **2**

(i) No odd numbers are equivalent to all 4 numbers are even. Even numbers are 2, 4, 6, 8. So only 1 way or 4C_4 way.
 (ii) at least one odd numeral = Total ways - no odd = ${}^9C_4 - 1 = 125$
 (iii) 3 odd or 4 odd = ${}^5C_3 \times {}^4C_1 + {}^5C_4 = 45$

B12	HURLSTONE	☒	☐
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An ant is to crawl along the black lines in the grid below to get from corner A to corner B. However, the ant can only travel along the lines either towards the right or in a downwards direction on the page. How many different routes from A to B are possible? **1**

Answer = 20
 (This is a version of organising outcomes from a diagram.)
 By tracing paths, there's:
 1 passing through C
 Another 3 passing through D
 Another 6 passing through E
 $1+3+6 = 10$ represents half the pathways from A, so answer = 20.



B13	JAMES RUSE	☒	☐
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A team of six students is to be formed from a class of ten students. How many different teams can be formed if two particular students cannot both be selected for the team? **2**

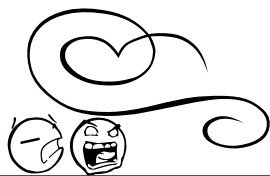
- A. 252 B. 210 C. 168 D. 140

Justification for Q4: Students: X X X X X X X A B
 If neither A or B: 28 possibilities ($8C6$ from non A and B)
 If A or B: 112 possibilities ($8C5 \times 2$)

B14	JAMES RUSE	☒	☐
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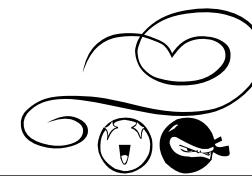
From seven girls and five boys, a committee of seven is to be chosen. What is the probability of choosing a committee containing at least four girls? **2**

Total = $N(4 \text{ girls}) + N(5 \text{ girls}) + N(6 \text{ girls}) + N(7 \text{ girls})$
 $= ({}^7C_4 \times {}^5C_3) + ({}^7C_5 \times {}^5C_2) + ({}^7C_6 \times {}^5C_1) + ({}^7C_7 \times {}^5C_0)$
 $= 350 + 210 + 35 + 1 = 596$
 $N(\text{no restriction}) = {}^{12}C_7 = 792$
 $\therefore \text{Probability}(\geq 4 \text{ girls}) = \frac{596}{792} = \frac{149}{198}$



combinatorics

Part B : combinations



B15	KNOX	☒	☐
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Consider the digits 7, 6, 5, 4, 3, 2, 1 and 0. Find how many four-digit numbers are possible if the digits are in:

- descending order,
- ascending order.

(b) 7, 6, 5, 4, 3, 2, 1 & 0
selecting 4 such that digits are in
(i) Descending order: selecting any 4 digits & then they can be arranged in the required order. ${}^8C_4 = 70$
(ii) Ascending order: cannot start with 0 so, ${}^7C_4 = 35$

1
1

B16	PEM	☒	☐
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A group of 40 new Year 7 students is going to be randomly divided into two classes of 20 students. Four of the students, Anne, Ben, Carrie and David have been together in the same class since they started school and are close friends.

- What is the probability that the four friends will be in the same class? Write your answer correct to 4 decimal places.
- What is the probability that exactly three of the friends will be in the same class? Write your answer correct to 4 decimal places.

2
2

- (a) (2 marks) Put 4 friends in class A, then need to choose 16 more students from 36
Sample answer: Probability all four friends in class A = $\frac{{}^{36}C_{16}}{{}^{40}C_{20}}$
Could also all be in class B
Probability all in same class = $2 \times \frac{{}^{36}C_{16}}{{}^{40}C_{20}} = 0.106029... = 0.106$ (to 3 decimal places)
- (b) (2 marks) Put 3 friends in class A. The 3 friends can be chosen in 4C_3 ways then need to choose 17 more students from 36 (because 4th friend can't be in class A)
Sample answer: Probability exactly 3 friends in class A = $\frac{{}^4C_3 \times {}^{36}C_{17}}{{}^{40}C_{20}}$
Could also all be in class B
Probability exactly 3 friends in same class = $2 \times \frac{{}^4C_3 \times {}^{36}C_{17}}{{}^{40}C_{20}} = 0.49896... = 0.499$ (to 3 decimal places)

B17	PEM	☒	☐
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From a group of twenty senators, five will be chosen to form a committee.

1

One senator will be chairperson of the committee, another will be secretary of the committee and the others will serve as members. In how many ways can the committee be formed?

(A) ${}^{20}C_2 \times {}^{18}C_3 \times 5!$

(C) ${}^{20}C_1 \times {}^{19}C_1 \times {}^{18}C_3$

(B) $({}^{20}C_1 + {}^{19}C_1) \times {}^{18}C_3$

(D) ${}^{20}C_1 + {}^{19}C_1 + {}^{18}C_3$

Chairperson chosen in ${}^{20}C_1$ ways.
Secretary chosen in ${}^{19}C_1$ ways.
Members chosen in ${}^{18}C_3$ ways.
 $\therefore$ number of ways = ${}^{20}C_1 \times {}^{19}C_1 \times {}^{18}C_3$

B18	PLC	☒	☐
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A garage has fifteen vehicles in it, of which five are vans and the remainder are sedans.

1

- In how many ways can a group of six vehicles be selected if there are no restrictions on the composition of the group?
- If the six vehicles are selected at random, what is the probability that the group of vehicles includes at least four vans?

(i) ${}^{15}C_6 = 5005$ (ii) ${}^5C_4 + {}^5C_5 = 5 + 1 = 6$
 $\therefore P(6) = \frac{6}{5005}$

B19	PLC	☒	☐
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A group of 40 new Year 7 students is going to be randomly divided into two classes of 20 students.

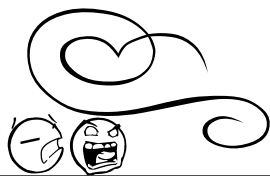
Four of the students, Anne, Ben, Carrie and David have been together in the same class since they started school and are close friends.

2

- What is the probability that the four friends will be in the same class? Write your answer correct to 3 decimal places.
- What is the probability that exactly three of the friends will be in the same class? Write your answer correct to 3 decimal places.

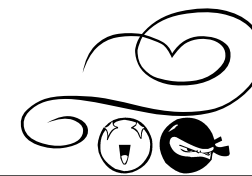
d. (i) $\frac{{}^4C_4 \times {}^{36}C_{16}}{{}^{40}C_{20}} = \frac{1 \times 93025}{108507}$
 $= 0.106029... = 0.106$
Divide into 2 classes and consider two groups of the 40.
Group the 4 students
OTHERS
The remaining

(ii) class A or class B.
 $\frac{{}^4C_3 \times {}^{36}C_{17} + {}^4C_3 \times {}^{36}C_{17}}{{}^{40}C_{20}} = \frac{2 \times 4C_3 \times 36C_{17}}{{}^{40}C_{20}}$
 $= 0.499$



combinatorics

Part B : combinations



B20	SYDNEY GIRLS	☒	☐
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A debating group consists of 12 students, 8 of whom are girls.

How many debating teams of three students can be formed if:

- There are no girls at all.
- There is exactly one girl.
- There is at least one girl.

$$\begin{aligned} (i) \quad & {}^8C_3 = 8 \\ (ii) \quad & 8 \times {}^7C_2 = 48 \\ (iii) \quad & 12C_3 - 4 = 216 \end{aligned}$$

1
1

B21	SYDNEY GRAMMAR	☒	☐
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A committee of four people is to be chosen from a group of eight women and five men.
How many committees are possible that consist of two women and two men?

2

$${}^8C_2 \times {}^5C_2 = 280$$

B22	TRIAL ENTERPRISES	☒	☐
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In how many ways can a soccer team of eleven be chosen from 15 players?

1

- (A) 1365 (B) 165 (C) 200 (D) 15

B23	TRIAL ENTERPRISES	☒	☐
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Two lines are parallel. On the first line there are 4 dots, and on the second there are 6 dots.

3

How many possible triangles can be formed by joining 3 of these dots?

Triangles are formed by selecting 2 dots from the upper line and 1 dot from the lower line i.e. ${}^4C_2 \times {}^6C_1 = 36$ triangles or by selecting 2 dots from the lower line and 1 dot from the upper line i.e. ${}^6C_2 \times {}^4C_1 = 60$ triangles. $\therefore$ total triangles is $36+60=96$

B24	TRIAL ENTERPRISES	☒	☐
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Committees of 6 persons are to be obtained from a group of 7 men and 5 women. How many committees can be formed if it includes a majority of men?

2

$$\begin{aligned} 6M \text{ and } 0W &= {}^7C_6 = 7 \\ 5M \text{ and } 1W &= {}^7C_5 \times {}^5C_1 = 105 \\ 4M \text{ and } 2W &= {}^7C_4 \times {}^5C_2 = 350 \\ \text{Total} &= 462 \end{aligned}$$

B17	PEM	☒	☐
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From a group of twenty senators, five will be chosen to form a committee.

1

One senator will be chairperson of the committee, another will be secretary of the committee and the others will serve as members. In how many ways can the committee be formed?

(A) ${}^{20}C_2 \times {}^{18}C_3 \times 5!$

(C) ${}^{20}C_1 \times {}^{19}C_1 \times {}^{18}C_3$

(B) $({}^{20}C_1 + {}^{19}C_1) \times {}^{18}C_3$

(D) ${}^{20}C_1 + {}^{19}C_1 + {}^{18}C_3$

Chairperson chosen in ${}^{20}C_1$ ways.
Secretary chosen in ${}^{19}C_1$ ways.
Members chosen in ${}^{18}C_3$ ways.
 $\therefore$ number of ways = ${}^{20}C_1 \times {}^{19}C_1 \times {}^{18}C_3$

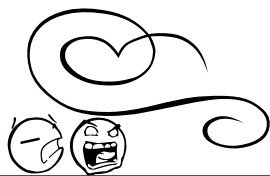
B18	PLC	☒	☐
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A garage has fifteen vehicles in it, of which five are vans and the remainder are sedans.

1

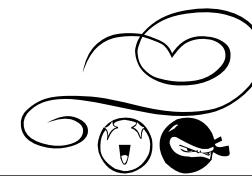
- In how many ways can a group of six vehicles be selected if there are no restrictions on the composition of the group?
- If the six vehicles are selected at random, what is the probability that the group of vehicles includes at least four vans?

$$\begin{aligned} (i) \quad & {}^{15}C_6 = 5005 \\ (ii) \quad & 4 \text{ or } 5 \\ & {}^5C_4 {}^{10}C_2 + {}^5C_5 {}^{10}C_1 = 235 \\ & \therefore P(G) = \frac{235}{5005} \end{aligned}$$



combinatorics

Part C : mixed



C1	BAULKHAM	☒	☐
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Express $\frac{{}^{10}C_k}{{}^9C_k}$ in simplest form

$$\frac{{}^{10}C_k}{{}^9C_k} = \frac{10!}{k!(10-k)!} \times \frac{k!(9-k)!}{9!}$$

$$= \frac{10}{(10-k)}$$

2

C2	BAULKHAM	☒	☐
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Which of the following is an expression for $\frac{1}{n!} - \frac{1}{(n+1)!}$?

1

- (A) $\frac{1}{n!(n+1)!}$ (B) $\frac{n}{n!(n+1)!}$ (C) $\frac{1}{(n+1)!}$ (D) $\frac{n}{(n+1)!}$

C3	CARINGBAH	☒	☐
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Three squares are chosen at random from the 3×3 grid below, and a cross is placed in each chosen square.

1



What is the probability that all three crosses lie in the same row, column or diagonal?

- (A) $\frac{1}{28}$ (B) $\frac{2}{21}$ (C) $\frac{1}{3}$ (D) $\frac{8}{9}$

9C_3 ways of selecting 3 squares.
There are 3 ways to select 3 crosses on the same row, 3 ways for the crosses to be on the same column, 2 ways for the diagonal = 8 total ways
Probability = $\frac{8}{{}^9C_3} = \frac{2}{21}$

C4	CARINGBAH	☒	☐
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A school music band consists of 8 dancers and a singer. The dancers are selected from 17 students in Year 9. The singer is selected from 5 students in Year 10. In how many ways could the band be selected?

1

- (A) ${}^{17}C_8 + {}^5C_1$ (B) ${}^{17}P_8 + {}^5P_1$ (C) ${}^{17}C_8 \times {}^5C_1$ (D) ${}^{17}P_8 \times {}^5P_1$

Choose 8 dancers from 17 dancers in ${}^{17}C_8$ ways.
Choose 1 singer from 5 singers in 5C_1 ways.
For each singer I chose, there are ${}^{17}C_8$ ways to choose the dancers.
Since there are 5C_1 ways to choose the singers, therefore total ways = ${}^{17}C_8 \times {}^5C_1$

C5	INDEPENDENT	☒	☐
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A group consists of 3 pairs of brothers and sisters from 3 different families. In how many ways can the group be seated in a circle so that the members of the same family are sitting opposite each other?

1

- (A) 8 (B) 12 (C) 16 (D) 20

Seat brother and sister from Family 1. Now relative to them, seat Family 2 in 4 ways then Family 3 in 2 ways. Hence 8 ways.

C6	KNOX	☒	☐
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A car hire company has 10 sports cars and 16 four-wheel drive cars. The company is to make a leaflet for a special offer. This leaflet is to contain 4 sports cars and 6 four-wheel drive cars. In how many different ways could the cars in the leaflet be chosen?

1

- A. ${}^{10}P_4 \times {}^{16}P_6$ B. ${}^{10}C_4 \times {}^{16}C_6$ C. ${}^{10}P_4 + {}^{16}P_6$ D. ${}^{10}C_4 + {}^{16}C_6$

Q3) There are ${}^{10}C_4$ ways to choose the 4 sports cars for the leaflet. Each of these combinations could be paired with any one of the ${}^{16}C_6$ combinations of 6 four-wheel drive cars. $\therefore {}^{10}C_4 \times {}^{16}C_6$ (B)

C7	KNOX	☒	☐
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In how many ways can 5 people from a group of 14 people be chosen and then arranged in a circle?

2

- A. $\frac{13!}{8!}$ B. $\frac{13!}{8!5}$ C. $\frac{14!}{9!}$ D. $\frac{14!}{9!5}$

$$\frac{14!}{5!9!} \times 4! = \frac{14!}{9!5}$$

C8	TOTAL EDUCATION	☒	☐
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Solve the equation ${}^{n-1}C_{n-3} + {}^{n-2}C_{n-3} = 20$. Show your working.

3

$$\frac{{}^{n-1}C_{n-3} + {}^{n-2}C_{n-3}}{(n-1)! + (n-2)!} = 20$$

$$\frac{2!(n-3)! + 1!(n-3)!}{(n-1)(n-2) + (n-2)} = 20$$

$$\frac{2(n-2) + 1}{n^2 - n - 42} = 20$$

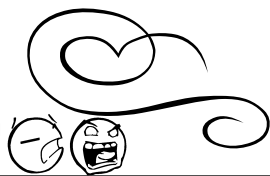
$$(n-1)(n-2) + 2(n-2) = 40$$

$$n^2 - n - 42 = 0$$

$$(n-7)(n+6) = 0$$

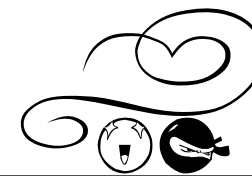
$$n = 7, -6$$

But $n \geq 3$ since NC_R requires $N, R \geq 0$ (and, of course, $N \geq R$).
 $\therefore n = 7$



combinatorics

Part C : mixed



C9	TOTAL EDUCATION	☒	☐
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In how many ways can eight differently coloured points be equally spaced around two identical circles if four points are to be placed around each circle? 1

- A. 36 B. 576 C. 1260 D. 2520

5. Consider choosing 4 coloured points for the first circle. This can be done in 8C_4 ways. These can then be arranged in a circle in $3!$ ways. The remaining 4 coloured points for the second circle can also be arranged in $3!$ ways. Finally, we divide by 2 as the circles can be identical so we need to eliminate duplicate arrangements.

$$\therefore \text{total arrangements} = \frac{{}^8C_4 \times 3! \times 3!}{2}$$

C10	TOTAL EDUCATION	☒	☐
Show that for all integers $n \geq 3$, $\binom{n+2}{3} - \binom{n}{3} = n^2$ L.H.S. = $\binom{n+2}{3} - \binom{n}{3}$ $= \frac{(n+2)!}{3!(n-1)!} - \frac{n!}{3!(n-3)!}$ For a common denominator, consider, $= \frac{(n+2)!}{3!(n-1)!} - \frac{n!}{3!(n-3)!} \times \frac{(n-1)(n-2)}{(n-1)(n-2)}$ $= \frac{(n+2)!}{3!(n-1)!} - \frac{(n-1)(n-2)n!}{3!(n-1)!}$ $= \frac{n![(n+2)(n+1) - (n-1)(n-2)]}{3!(n-1)!}$ $= \frac{n!(6n)}{3!(n-1)!}$ $= \frac{n(n-1)!(6n)}{3!(n-1)!}$ $= n^2, \text{ as required.}$			

C4	TOTAL EDUCATION	☒	☐
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A school music band consists of 8 dancers and a singer. The dancers are selected from 17 students in Year 9. 1
The singer is selected from 5 students in Year 10. In how many ways could the band be selected?

- (A) ${}^{17}C_8 + {}^5C_1$ (B) ${}^{17}P_8 + {}^5P_1$ (C) ${}^{17}C_8 \times {}^5C_1$ (D) ${}^{17}P_8 \times {}^5P_1$

Choose 8 dancers from 17 dancers in ${}^{17}C_8$ ways.
Choose 1 singer from 5 singers in 5C_1 ways.
For each singer I chose, there are ${}^{17}C_8$ ways to choose the dancers.
Since there are 5C_1 ways to choose the singers,
therefore total ways = ${}^{17}C_8 \times {}^5C_1$

C11	TOTAL EDUCATION	☒	☐
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The ten letters of the word AMALGAMATE are arranged at random in a row by a random generator.

- (i) What is the total number of different arrangements? 1
(ii) The random generator makes 10^6 arrangements. 2
If the arrangement ETAMAGLAMA appeared most commonly, what is the smallest number of times it could have appeared?
(iii) In how many different arrangements does the letter A appear immediately before the letter M? 2

Question 8(d) (ii)

If there are 75600 different arrangements (using part (i)) and the random generator makes 10^6 arrangements, the number of times each arrangement appears is:

$$\frac{10^6}{75600} \approx 13.22 \dots \text{ times}$$

Question 8(d)(i)

Sample answer:

Since there are 4 As and 2 Ms,

$$\frac{10!}{4! \times 2!} = 75600$$

Considering the pigeonhole principle, if each arrangement appears 13 times, we have $13 \times 75600 = 982800$.

Since the random generator makes $10^6 = 1\,000\,000$ arrangements, we still need to consider the remaining $1\,000\,000 - 982\,800 = 17\,200$ arrangements which need to be distributed to the 75600 different permutations. If we assign 2 of these to the word ETAMAGLAMA so that it appears most commonly, the smallest number is 15.

Question 8(d)(iii)

Sample answer:

If we group an A and an M together, we have 2 of these pairs. Together with the remaining 6 letters in the word AMALGAMATE, we consider an arrangement of 8 which can be made in $8!$ ways. However, we need to account for the 2 pairs of "AM" (duplicates) and the 2 remaining As.

$$\therefore \frac{8!}{2! \times 2!} = 10\,080 \text{ different arrangements.}$$